Experiment : 1

Date: 19-0

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

AIM

To study the concepts of Artificial Intelligence.

THEORY

1. What is AI?

Artificial Intelligence (AI) is a branch of computer science focused on building systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, perception, and problem-solving. Coined by John McCarthy in 1956, modern AI utilizes algorithms, large datasets, and computational power to make autonomous decisions.

2. What are the different sub-sets of AI?

- Narrow AI (Weak AI): Systems designed for specific, constrained tasks (e.g., facial recognition, spam filters).
- General AI (Strong AI): Theoretical systems possessing human-level versatility across all domains.
- Super AI: Hypothetical AI surpassing human intelligence in every aspect.
- Reactive Machines & Limited Memory: Functional subsets based on memory utilization and real-time response.

3. Explain any four real-world applications of AI

- Healthcare: AI analyzes medical imaging (X-rays, MRIs) for early disease detection and assists in drug discovery.
- Autonomous Vehicles: Self-driving cars (Tesla, Waymo) use computer vision and sensor fusion to navigate safely.
- Recommendation Systems: Netflix and Amazon analyze user behavior to suggest personalized content and products.
- Virtual Assistants: Siri, Alexa, and conversational agents use Natural Language Processing to interact and execute tasks.

4. What is the Turing test?

Proposed by Alan Turing in 1950, the Turing test evaluates a machine's ability to exhibit intelligent behavior indistinguishable from a human. During a blind text conversation, if a human judge cannot reliably distinguish the machine from a human participant, the machine passes.

5. What is the difference between Weak AI, Strong AI and Super AI?

- Weak AI (Narrow AI): Operates within a narrow predefined domain with no

consciousness (e.g., Siri, chess bots).

- Strong AI (General AI): Matches human cognitive flexibility across any intellectual task (currently theoretical).

- Super AI: Exceeds human intelligence and capabilities across all fields (hypothetical future stage).

6. Differentiate between Artificial Intelligence, Machine Learning, and Deep Learning

- Artificial Intelligence (AI): The overarching umbrella field aimed at creating intelligent machines.

- Machine Learning (ML): A subset of AI where algorithms learn patterns from data without explicit programming.

- Deep Learning (DL): A specialized subset of ML utilizing multi-layered artificial neural networks for complex feature extraction.

7. Discuss the advantages and limitations of Artificial Intelligence

- Advantages: High automation and 24/7 productivity, rapid processing of massive data, reduction of human error, and breakthrough capabilities in complex problem-solving.

- Limitations: High deployment and compute costs, heavy reliance on quality data (risk of bias), lack of true common sense or consciousness, and ethical/privacy concerns.

8. Case Study: ChatGPT as an AI-based Application

a) What problem does the application solve? ChatGPT solves the problem of bridging human intent and machine execution in natural language. It eliminates the friction of information retrieval, text composition, code generation, and summarization by providing instant, context-aware, conversational responses. It democratizes access to expert-level drafting, brainstorming, and tutoring across diverse domains.

b) Which AI technique(s) are used? ChatGPT leverages state-of-the-art Natural Language Processing (NLP) built on the Transformer architecture (Generative Pre-trained Transformer). It utilizes self-attention mechanisms to capture long-range dependencies in text, Supervised Fine-Tuning (SFT), and Reinforcement Learning from Human Feedback (RLHF) to align model outputs with helpful and safe human conversational norms.

c) What are the inputs and outputs of the system? The input is a user-defined natural language prompt (instructions, questions, code snippets, or conversational context). The output is a generated sequence of text tokens forming a coherent, contextually relevant, and human-like response or artifact.

d) What are its advantages? It offers 24/7 availability, immense versatility across writing, coding, analysis, and translation, rapid response times, significant boosts in individual productivity, and seamless integration capabilities via APIs for developers and enterprises.

e) What are its limitations? ChatGPT can occasionally generate incorrect, biased, or fabricated ('hallucinated') information with high confidence. It lacks genuine comprehension, emotional intelligence, and real-world sentience. Furthermore, it

poses risks regarding data privacy, misinformation, and intellectual property concerns.

CONCLUSION

In this experiment, we explored the foundational concepts of Artificial Intelligence, its subsets, real-world implementations, theoretical benchmarks like the Turing test, and structural distinctions between AI, ML, and DL. Through the detailed case study of ChatGPT, we analyzed how transformer-based models solve complex NLP tasks, their underlying techniques, capabilities, and inherent limitations. AI represents a transformative technological leap that requires careful ethical navigation and continuous oversight.